

Qualifying exam in Algebra. Fall 2022.

0. Write your name here:

Part I. Definitions and Theorems. (each problem 5 points)

1. State the universal property of localization of a commutative ring.

2. State the theorem on adjointness of $\otimes$ and Hom .

Part II. True or false. Give brief justification. (each problem 8 points)

1. Let $F, G : \mathcal{A} \rightarrow \mathcal{B}$ be isomorphic functors from category $\mathcal{A}$ to category $\mathcal{B}$, and $f, g \in \text{Hom}_{\mathcal{A}}(A, A')$. Then $F(f) = F(g)$ if and only if $G(f) = G(g)$.

2. Let $F = \mathbb{Q}(\sqrt[3]{2})$. Then for any $n \in \mathbb{Z}_{\geq 1}$, the degree $[F(\sqrt[n]{1}) : F] = \phi(n)$ (here $\phi(n)$ is the Euler function).

3. The embedding functor from finite abelian groups to all abelian groups admits a right adjoint.

4. Every group of order 2022 is solvable.

5. For a ring R , any finitely generated R -module has a simple quotient.

Part III. Longer problems. (each problem 10 points)

You have to state precisely any standard result that you are using.

1. Let P be a square matrix over $\mathbb{Q}$. Prove that $P^2 = P$ if and only if $\text{rank}(P) = \text{tr}(P)$ and $\text{rank}(\text{Id} - P) = \text{tr}(\text{Id} - P)$ (here Id is the identity matrix).

2. Prove that the ring $\mathbb{Z}[\sqrt{7}]$ is integrally closed.

3. Let χ and ϕ be some irreducible complex characters of a finite group G with $\chi(1) \neq \phi(1)$. Let $\chi\phi = \sum_i \psi_i$ be the decomposition of product character $\chi\phi$ into sum of (not necessarily distinct) irreducible characters ψ_i . Prove that none of the characters ψ_i is linear, i.e. $\psi_i(1) \neq 1$ for all i .

4. Assume an algebraic set $\mathcal{V}$ is a disjoint union of two non-empty closed subsets $\mathcal{V}_1$ and $\mathcal{V}_2$. Prove that the coordinate algebra of $\mathcal{V}$ contains a nontrivial idempotent.

5. Let A and B be two division algebras. Prove that A is Morita equivalent to B if and only if $A \simeq B$.